

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: Bai Zi Jun

Student ID: 22097967d

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

 show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 128

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

 R2 is advertising routes only toward R1, not toward R3. Therefore, R3 learns the 192.168.2.0/24 network indirectly through R1, so the path is R3 to R1 to R2. Since each serial link cost 64, the accumulated cost is $64 * 2 = 128$

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

 router ospf 1

 no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

 65 It is calculated as 64 (the cost of R3's serial link to R2) + 1 (the cost of R2's G0/0 interface to 192.168.2.0/24)

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because the default OSPF reference bandwidth is only 100 Mb/s. With that default, Fast Ethernet, Gigabit Ethernet, and even faster links can end up with the same cost of 1, so OSPF cannot reflect real link-speed differences accurately. Raising the reference bandwidth makes route costs more realistic.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to **192.168.3.0/24** is **782**, calculated as **781** for R1's serial link plus **1** for R3's G0/0 interface. The cost to **192.168.23.0/30** is **845**, calculated as **781 + 64** across the two serial links on the path. At that point there are two equal-cost paths to 192.168.23.0/30, one through S0/0/0 and one through S0/0/1.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562, because after setting all serial interfaces to bandwidth 128, each serial link has a cost of 781, so the total cost to 192.168.23.0/30 becomes $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because the cost on R1's S0/0/1 interface was manually changed to **1565**, which makes the direct path to R3 more expensive than the route through R2. The path through R2 has an accumulated cost of **1562**, so OSPF chooses that lower-cost route.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The router ID uniquely identifies each router in the OSPF domain. The lab also shows that if you do not set it manually, Cisco chooses it by precedence: first router-id, then highest loopback IP, then highest active physical-interface IP. Controlling it makes identification stable and predictable, which helps with neighbor relationships, troubleshooting, and avoiding confusion when interface addresses change.

2. Why is the DR/BDR election process not a concern in this lab?

Because the routers form their OSPF adjacencies mainly over **point-to-point serial links**, and DR/BDR election is not used on point-to-point OSPF networks. The serial interfaces in the lab are shown as POINT_TO_POINT, so DR/BDR is not relevant to the main routing adjacencies here.

3. Why would you want to set an OSPF interface to passive?

To stop sending OSPF Hello packets and routing updates on interfaces where no OSPF neighbor is needed, especially LAN interfaces connected only to end hosts. This reduces unnecessary routing traffic while still allowing the connected network to be advertised.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF organizes routers into **areas**, with Area 0 as the backbone, which makes the protocol hierarchical and scalable. The lab itself is a single-area OSPFv2 lab and shows the backbone area in the OSPF output.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

OSPF